

The Evidence: What We Proved in Part 2

Part 1 was the theory. Part 2 was the proof. We ran **1,594 tests** against **950 attack samples** across **four production architectures**.

Here is what the data says.

The Experimental Setup

We tested four common multi-agent architectures found in production:

- ▶ **Claude Code** (Hierarchical)
- ▶ **AutoGPT** (Autonomous Loop)
- ▶ **CrewAI** (Role-Based Team)
- ▶ **LangGraph** (State Machine)

The test corpus included direct prompt injection, poisoned RAG contexts, deep trust exploitation, and multi-turn social engineering.

Finding 1: Defense Layering vs. Individual Efficacy

The Data: Individual defenses (like just a firewall) stopped ~60-70% of attacks. The full CIF stack stopped **97%**. **The**

Implication: The defenses demonstrated orthogonal coverage

A Note on the Numbers

The detection rates in Part 2 are derived from a calibrated parametric simulation, modeled on the architecture's topology. They represent the *structural* security of the design.

- ▶ **97% Detection** means: "In a fully implemented system of this topology, 97% of these attack vectors violate a defined constraint."
- ▶ It does **not** mean: "We have a magic Python script that catches 97% of all evil AI thoughts."

We proved the *architecture* works. The implementation fidelity is the variable for the builder.

Tripwire Configuration Data

The simulations utilized the following tripwire densities to achieve the reported results:

Category	Count Used in Sim	Placement Strategy
Identity canaries	3+ per agent	Core identity beliefs
Boundary canaries	5+ per agent	Permission boundaries
Principal canaries	2+ per agent	Trust relationships